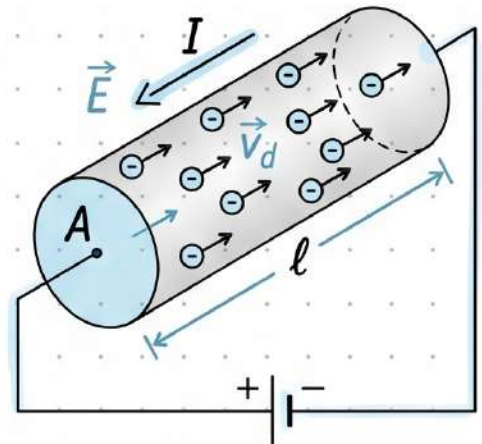


Current Elec: Fundamentals

Electric Current Basics

Instantaneous current: $I = \lim_{\Delta t \rightarrow 0} \frac{\Delta q}{\Delta t} = \frac{dq}{dt}$

Current in a conductor: $I = nqAv_d$
(where n is charge carriers per unit volume, A is area, v_d is drift velocity)



Drift Velocity & Mobility

Drift velocity: $\vec{v}_d = -\frac{e\vec{E}}{m}\tau$ (where e is electron charge, m is mass, τ is relaxation time)

Mobility: $\mu = \frac{|v_d|}{E} = \frac{e\tau}{m}$ with SI unit m^2/Vs

Current Density: $\vec{J} = \frac{ne^2}{m}\tau\vec{E}$ and $I = \int \vec{J} \cdot d\vec{A}$

Ohm's Law & Resistance

Ohm's Law: $V = IR$

Resistivity (specific resistance): $\rho = \frac{m}{ne^2\tau} = \frac{1}{\sigma}$, where σ is conductivity

Temperature dependence: $R = R_0(1 + \alpha(T - T_0))$, where α is the thermal coefficient of resistivity

Power & Heat

Electrical Power: $P = VI = I^2R = \frac{V^2}{R}$

Heat: $H = VIt = I^2Rt = \frac{V^2}{R}t$

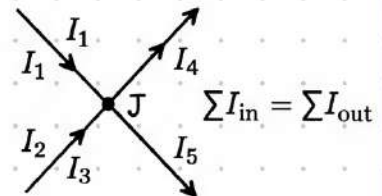
Kirchhoff's Laws & Grouping

Kirchhoff's Laws:



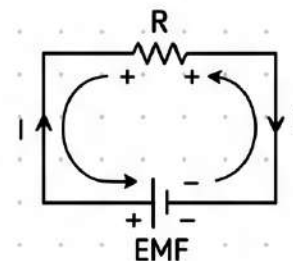
I. Law (Junction Law / KCL):

- Based on the law of conservation of charge.
- The algebraic sum of the currents meeting at a junction is zero ($\sum I_{in} = \sum I_{out}$).



II. Law (Loop Law / KVL):

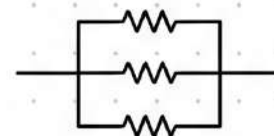
- The algebraic sum of all the voltages in a closed circuit is zero ($\sum IR + \sum EMF = 0$).
- While traversing a loop, if a higher potential point is entered, put a +ve sign, or if a lower potential point is entered, put a negative sign.



Combination of Resistances:

$$R_{eq} = R_1 + R_2 + R_3 + \dots + R_n$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$



Grouping of Cells:

• Series:

$$E_{eq} = E_1 + E_2 + \dots + E_n \text{ (use emf with polarity)}$$

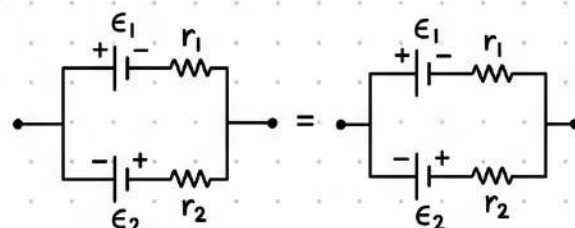
$$r_{eq} = r_1 + r_2 + r_3 + \dots$$



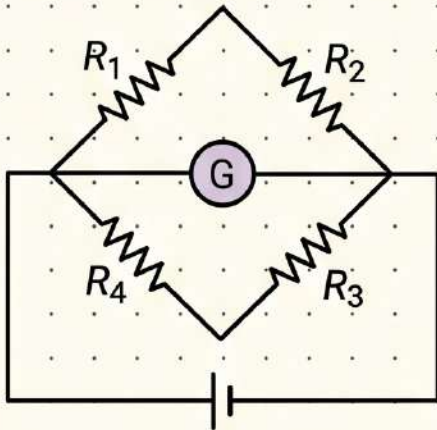
• Parallel:

$$E_{eq} = \frac{\frac{\epsilon_1}{r_1} + \frac{\epsilon_2}{r_2} + \dots}{\frac{1}{r_1} + \frac{1}{r_2} + \dots}$$

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \dots$$

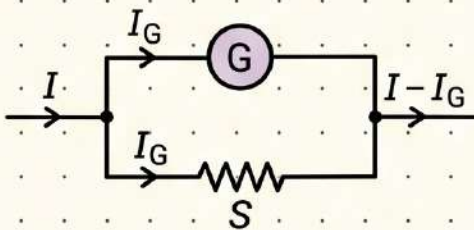


Instruments & Bridges



→ **Wheatstone Network:** Current through the central galvanometer resistance is zero (null point or balance point) if $\frac{R_1}{R_4} = \frac{R_2}{R_3}$.

→ **Ammeter:** A shunt (small resistance) is connected in parallel with a galvanometer to convert it into an ammeter. An ideal ammeter has zero resistance.

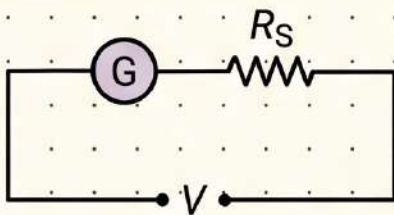


Shunt for the formula is

$$S = \frac{I_G \cdot R_G}{I - I_G}$$

If $I \gg I_G$, then $S = \frac{I_G \times R_G}{I}$.

→ **Voltmeter:** A high resistance is put in series with a galvanometer.

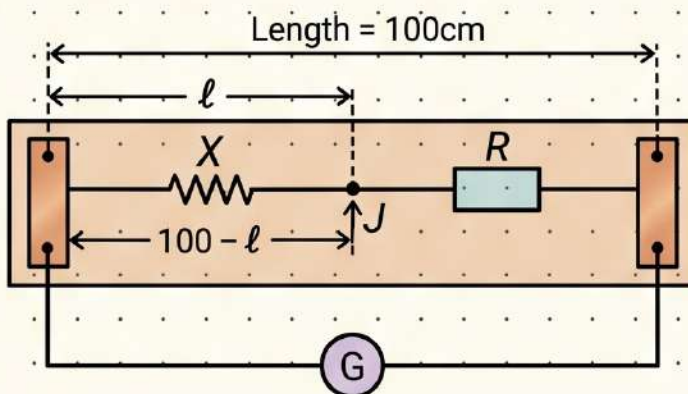


Series resistance for maximum potential difference.

$$V = I_G(R_S + R_G)$$

If $R_G \ll R_S$, then $R_S \approx \frac{V}{I_G}$.

→ **Metre Bridge:** It is used to measure unknown resistance.



Applying the condition for a balanced Wheatstone bridge, the unknown resistance is

$$X = \frac{100 - l}{l} R$$

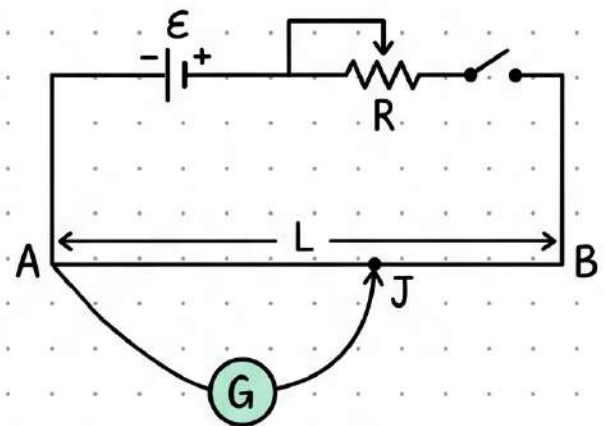
Potentiometer Applications

Potentiometer Principle

Potential gradient (x) is the potential difference per unit length of wire.

$$x = \frac{V_A - V_B}{L} = \frac{\epsilon}{R + r} \cdot \frac{R}{L}$$

Potentiometer is an ideal voltmeter because it does not draw any current from the circuit at the balance point.



Comparing EMFs

To find the EMF comparison we find at the emf of an unknown cell and compare the forms of the emf of two cells.

$$\frac{\epsilon_1}{\epsilon_2} = \frac{l_1}{l_2} \quad (\text{where } l_1 \text{ and } l_2 \text{ are balance lengths})$$

Finding Internal Resistance

Using internal resistance r' can be balance lengths for an open and closed secondary circuit in formula:

$$r' = \left[\frac{l_1 - l_2}{l_2} \right] R \quad (\text{where } R \text{ is known resistance})$$